

Balancing Data Sovereignty and Biodiversity Preservation in Automated Camera Trap Analysis

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Abstract

As machine learning (ML) systems increasingly demand unprecedented volumes of data, the issue of data sharing becomes pivotal. This is especially true in biodiversity research, where sensitive data and concerns over data sovereignty often hinder global cooperation. Our project addresses the nuanced tradeoffs associated with data sharing in ecological applications of ML.

By integrating data from multiple sites, ML models can enhance species detection and ecological monitoring. However, our research uncovers complexities in data sharing, particularly regarding long-tailed data distributions and species rarity. We propose strategies to optimize data sharing for ecological ML applications while considering site-specific factors and model performance tradeoffs, thereby promoting effective wildlife preservation amidst climate change and ecological degradation.

A. The Evolving Landscape of CV in Ecology

With the growth of the field of ML, there has been a corresponding surge in its applications across various domains, including ecology. The increasing adoption of ML in ecological research has opened up new avenues for understanding and conserving ecosystems. Large-scale ML models have been successfully applied in diverse ecological contexts, ranging from bat acoustics [3] to pollen analysis [9] and animal pose estimation [11]. Reviews such as those by Borowiec et al. [5] and Beery et al. [4] have highlighted the intersections of ML and ecology research, showcasing the transformative potential of ML in advancing ecological understanding.

One particularly impactful application of ML in ecology is the automation of camera trap data processing for wildlife monitoring. Camera traps, motion- or heat-

triggered cameras, are deployed extensively in ecological studies to capture images of wildlife, facilitating systematic observations of animal populations [1, 8]. These devices generate vast amounts of data and offer insights into species abundance, distribution, and behavior across diverse habitats.

Until recently, extracting information from camera trap images was a laborious and time-intensive process, reliant on the manual identification and labeling of countless animal photographs. However, the advent of computer vision (CV) technologies has revolutionized this process; ML models trained on camera trap data can automatically analyze and classify images with high accuracy and efficiency [16]. This transformation has streamlined wildlife monitoring processes, facilitating data-driven conservation efforts.

Innovations like these have propelled the evolution of ecology into a data-intensive discipline [3–5, 9, 11], spurred by the urgent need to monitor and safeguard global biodiversity. However, in this application of ML just as many others, the gold standard of aggregating and sharing as much data as possible holds true, and the effectiveness of ecology as a field has proceeded to hinge, more and more, on the quality and diversity of training data available—a fact which introduces problems unconquered even by the latest of CV technologies. For instance, camera trap datasets present many inherent challenges. The species in these datasets often follow a long-tailed distribution, with some animals being much more uncommon than others; this poses challenges for ML models trained on insufficient and/or imbalanced data. Data collected without human intervention may also feature partially occluded, blurry, or poorly lit species, further complicating model training. Moreover, species observation frequency fundamentally varies across time and space due to natural migration, habitat suitability, and anthropogenic influences [7]. Thus, the nature of camera trapping itself is an obstacle to obtaining sufficient labeled data for training robust models.

Data sharing among ecological researchers presents

a potential solution to these challenges, allowing for the pooling of datasets from multiple sources to create larger, more diverse training sets. Yet, despite the potential benefits of data sharing, there are greater nuances to consider. Understanding how different data sharing strategies impact model performance is essential for optimizing ecological ML applications.

B. Ecological Data Sharing and its Challenges

Arguments against the justification of mass data collection and aggregation within biodiversity research present unique considerations that differ from those in more predominant fields. While in other contexts, concerns about data scraping often revolve around unethical practices or profit-driven motives, the stakes in biodiversity research are much higher—essentially a matter of saving the world.

Local parks and governments responsible for collecting and holding wildlife camera trap images have valid reasons for wanting to keep their data private. Firstly, images of endangered animals could contain information on their location [1] and—if wrongly acquired by poachers, for example—end up abetting their eventual demise. Data contributors also raise concerns about fairness and disproportionate gain, i.e., not receiving adequate benefits or attribution in return for sharing their valuable wildlife data [6, 15]. Finally are concerns surrounding data autonomy. Data centers processing biodiversity data are usually located in more economically developed regions, e.g., Western Europe, but the places they want data from are in Africa, South America, etc., i.e., regions with greater racial diversity and less wealth. The pursuit of data from underrepresented regions by ML researchers coincides with heightened pressure on relevant communities to divulge sensitive data [12]. Sharing aside, acquiring diverse and well-labeled data poses its own set of challenges, often requiring domain experts and significant financial resources [13]. The logistics of transferring large-scale datasets, particularly in remote or underdeveloped regions with limited bandwidth, present additional hurdles.

It is essential to address these concerns ethically and practically. Despite ML’s heavy reliance on high-quality data, the implications of data sharing on model performance remain relatively unexplored within the ecological research community. While more data generally correlates with improved performance, nuanced

considerations, such as subgroup performance and data diversity, must be taken into account. As ecological research continues to embrace data-driven approaches, it is imperative to critically examine the trade-offs associated with data sharing, ensuring that ecological models are robust, equitable, and effective in advancing biodiversity conservation efforts. Thus, our research aims to shed light on the optimal strategies for sharing ecological data while respecting local concerns and data sovereignty, ultimately advancing the development of ML models for biodiversity analysis.

C. Related Work

Previous research in ecological ML has primarily focused on developing and evaluating ML models for species detection and monitoring using camera trap data. Various techniques, including convolutional neural networks and transfer learning, have been employed to train models capable of accurately identifying wildlife species from camera trap images.

One key challenge in ecological ML is the long-tailed distribution of species in camera trap datasets. Many species are rare or infrequently observed, leading to imbalanced training data that can negatively impact model performance. Previous studies have explored techniques for mitigating the effects of imbalanced data, such as oversampling rare species or incorporating class weights into the training process.

Another area of research relevant to our project is the impact of data sharing on model performance. While data sharing can potentially improve model generalization by increasing the diversity of the training set, it also introduces challenges related to data quality, compatibility, distribution, and bias. Understanding how different data sharing strategies affect model performance is essential for optimizing ecological ML applications.

Our research builds upon existing methodologies by delving deeper into the nuances of data sharing in ecological ML. By conducting a systematic analysis of the performance of ML models trained using different data sharing strategies, we aim to provide actionable insights for ecological researchers seeking to leverage ML techniques for species detection and monitoring.

D. Case Study Introduction

We explore the potential for regional adaptation of models, where only a portion of the data would need

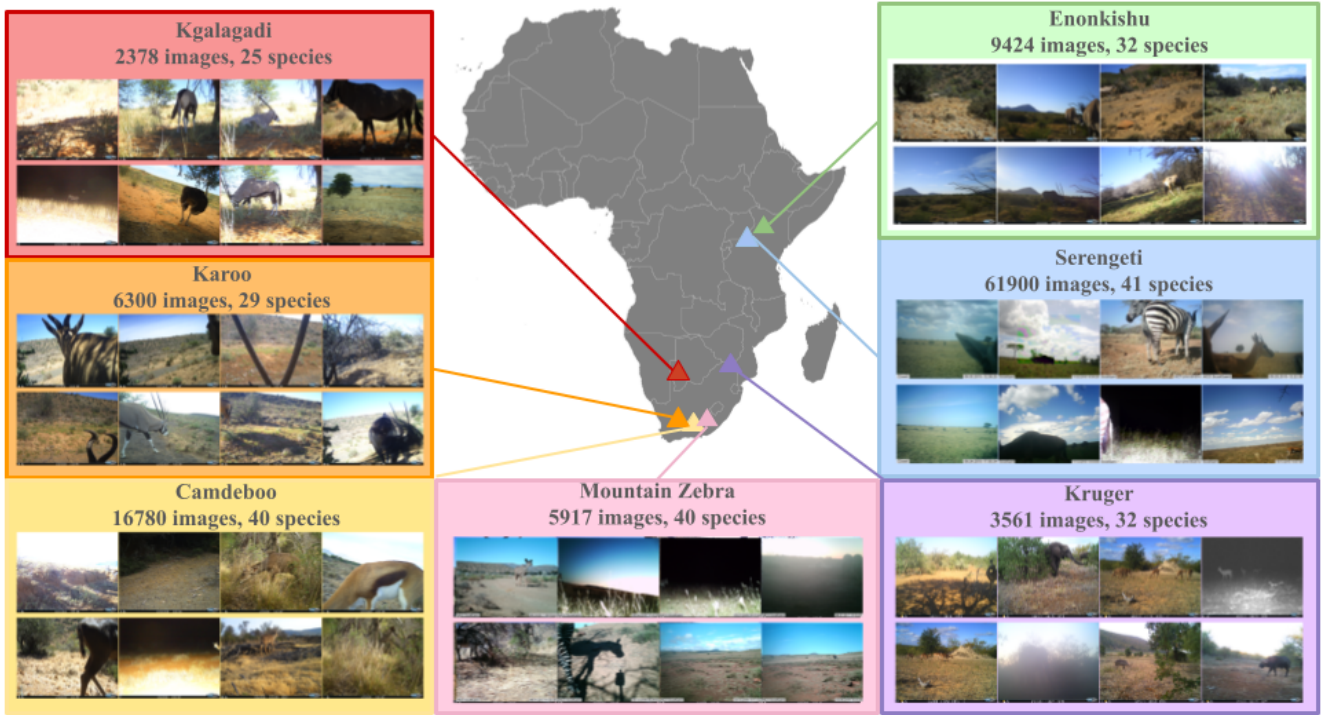


Figure 1. Ecological monitoring data from protected areas have long-tailed distributions over species of interest that are often quite different from each other due to differing habitat suitability and management practices, which, paired with vastly different available resources, technological capacity, and nuanced considerations when publishing or sharing data, lead to unequal access to well-performing machine learning models. We provide a systematic case study of realistic ecological data sharing scenarios built from camera trap data collected as part of the Snapshot Safari network, focusing on seven protected sites across Africa (locations and data samples shown).

to be shared. This selective sharing could potentially promote wildlife preservation just as effectively as their potentially overproduced counterparts. For instance, rather than collecting thousands of photos of the same species from various regions, a few carefully selected samples from specific regions may suffice to create accurate models for particular purposes globally. The fundamental premise of ML is to capture patterns in data that generalize well to unseen instances. In the context of wildlife identification, it is crucial to discern whether amassing vast quantities of data from a single class truly enhances model performance or if a more strategic approach could yield comparable results.

Beyond the level of individual species, we also question whether it is necessary to amass images from “everywhere,” including across biomes, even if those biomes hold completely different types of wildlife. Realistically, models are unlikely to be applied universally across entire continents or regions. Instead, they may be tailored to specific ecosystems or conservation areas, where the species composition and environmental

factors are relatively consistent.

Moreover, there is a need to evaluate the extent to which rare instances influence model efficiency, considering that species rarity and distribution heterogeneity pose significant obstacles to automated processing (in addition to increasing risk of endangerment).¹

We answer the above questions quantitatively through a case study on the Labeled Information Library of Alexandria: Biology and Conservation (LILA BC) Snapshot Safari dataset network, encompassing camera trap data from multiple national parks in Africa and labeled by citizen scientists via Zooniverse [14]. Through quantitative analysis and performance evaluation metrics, we aim to provide actionable insights

¹In practice, defining thresholds for what constitutes “enough” data is challenging. For instance, in our experiments, we define rare species as those with fewer than 100 examples, as per the *rough* (even less consistent when samples are not independent, as with camera trap data) rule of thumb that 100 examples should be “enough” to learn from. This number was *not* chosen based on any ecological information (e.g., as defined by the IUCN Red List for endangered species).

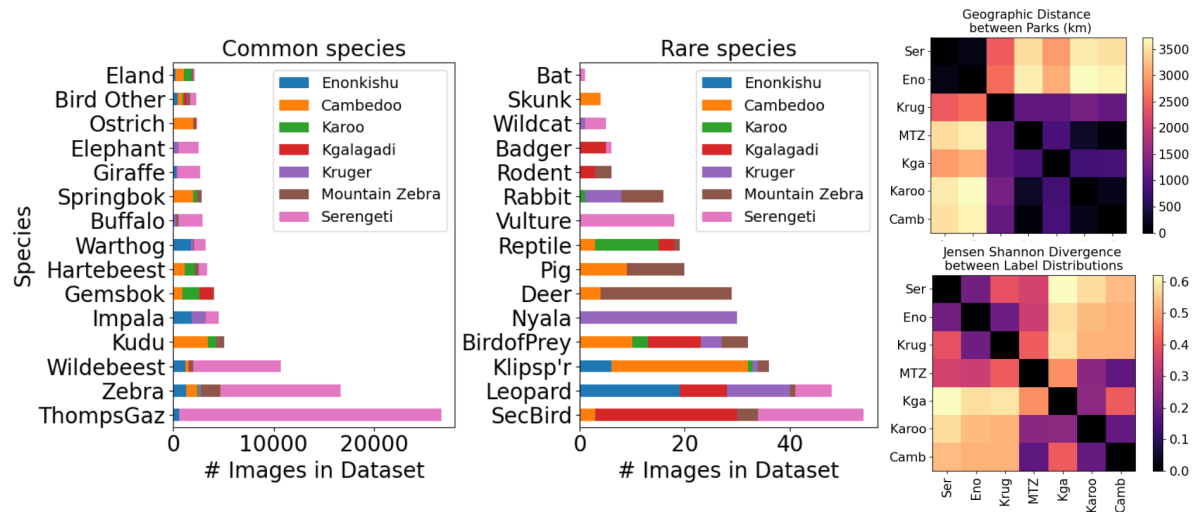


Figure 2. **Different sites have notably different distributions, both geospatially and across species.** Note that the distribution over species is long tailed and imbalanced, and quite different across sites. The pairwise geographic distance (top right) and the Jensen Shannon Divergence between the label distributions of each site pair (bottom right), which capture local similarity and structure.

that could inform data sharing policies and practices in ecological research and conservation initiatives.

E. Methodology

We hypothesize that data sharing strategies tailored to the specific characteristics of ecological sites can improve the performance of ML models for species detection. To test this hypothesis, we conduct experiments using datasets comprising camera trap images from seven sites representing a variety of habitats, species, site management styles, and area sizes within Africa. (We also aggregate metadata in a standardized manner for site comparisons.) Figure 1 provides a map of the sites and example images for each. Figure 2 summarizes dataset statistics and distribution.

We train ML models on species identification using different realistic data sharing scenarios/strategies, including uniform sampling, site-specific sampling, and species-specific sampling. Our methodology ensures that the models are trained and evaluated on diverse ecological datasets, capturing the variability in species composition and environmental conditions across different sites. The performance metrics are carefully chosen to assess both the overall accuracy and the ability of the models to detect rare species, which are of particular interest for conservation efforts.

F. Experiments

In our experiments, we aim to elucidate the impact of local and global data sharing on ML performance within the constraints of ecological organizations or protected areas (sites). We delineate three pivotal factors influencing ML efficacy in this context and design a series of experiments accordingly.

The first factor, **access to shared models**, assesses the performance implications of utilizing pre-trained models trained on data which may or may not share similar distributions with one’s own. We simulate scenarios where users have no ability to train new models but have access to “global” models trained on data from remaining sites (Cambedoo, Karoo, Serengeti, Mountain Zebra, and Kruger), as well as “local” models trained on the nearest geographical site (Serengeti for Enonkishu and Karoo for Kgalagadi [Figure 2]).

The second factor, **access to ML capacity and labelled data**, drives scenarios where users possess the resources (e.g., GPUs) and expertise to train custom models using a combination of global and/or local data (through shared published datasets or data sharing agreements) with in-house site-specific labelled data.

The third factor, **access to expert knowledge**, examines the influence of domain-specific knowledge (e.g., expected species list *or* species distributions/estimated relative frequency) about a site on ML performance. We explore two approaches: one without retraining,

utilizing Bayesian priors based on species presence or distribution at evaluation,² and another involving re-training the model by adjusting the training data based on species lists or distributions.³

For our experiments, we select Enonkishu and Kgalagadi as hypothetical users (test sites) due to their high species diversity, representing the eastern and southern clusters of the Snapshot network. We simulate a common deployment scenario where a site collects unlabeled data over a season and intends to employ ML for labeling. We randomly select 20% of each site’s data (~1000 images) to represent potential at-site labelled data, with the remainder serving as the test set.

All models are trained with a ResNet50 backbone, pretrained on ImageNet weights, and fine-tuned using training set data. The experiments are conducted on cropped regions of camera trap images corresponding to MegaDetector animal detections [4],⁴ reducing the impact of repeated backgrounds. Additionally, our experiments utilize a batch size of 64, learning rate of 0.001, weight decay of 0.01, and random seed 42.

G. Analysis

The results of our experiments, detailed in Tables 1 and Appendix Table 2, highlight several key insights:

Challenging Assumptions in ML: Our findings challenge the conventional wisdom that larger training datasets invariably lead to improved model performance. Our experiments reveal that, without additional

²...by reweighing the model’s likelihoods with an approximated Bayesian prior based on species presence or frequency [2]. While, in practice, the assumption that site location and image features are conditionally independent (given class label) is not strictly true, it affords this remapping of Bayes’ rule (I : image; z : corresponding site; y : class label):

$$P(y|I, z) = \frac{P(I, z|y)P(y)}{P(I|z)} \approx P(y|I)P(y|z)$$

A list of known site species could then be used to estimate $P(y|z)$ as a uniform distribution over the known species (0 everywhere else). Relative species frequencies could, even more ideally, be mapped to a normalized distribution over known species as an estimate of $P(y|z)$. We use an oracle to estimate this distribution, matching the site’s real aggregate distribution and, thus, providing an upper bound on the potential benefit of domain knowledge. Detailed results are presented in Table 1.

³With only access to a species list, we eliminate any species not present in the test site from our training set. We then under/oversample training data to match the test site distribution, keeping the size of the training dataset fixed (results in Appendix Table 2).

⁴Available along with our case study data on the LILA data repository [10].

domain knowledge, local models trained on smaller datasets outperform global models trained on larger datasets. For instance, on Enonkishu and Kgalagadi, global models without site-specific knowledge led to decreases in accuracy and recall by 17.6% and 4.5%, and 3.5% and 10.5%, respectively (Table 1). Without site-specific knowledge, adding data from further afield (naturally having more distinct training distributions [Figure 2] influenced by increasingly different habitat characteristics) biases the model away from a given site’s categories and distribution.

Value of Local Data: By conducting a data source ablation (Appendix Figure 5) comparing downstream performance on Enonkishu with three different data sources (all sites, all sites minus local (Serengeti), or local), we further observed how incorporating data from nearby sites often increases performance. However, the effect is highly species-dependent; adding data sources may confuse the model, leading to a suppression in performance for certain species. This underscores the need for a nuanced approach to data aggregation, tailored to the specific use case and species distribution.

Benefit of Domain Knowledge: Even in the absence of ML capacity for on-site training, incorporating site-specific domain knowledge, such as species lists or distributions, significantly enhances shared model performance. Bayesian remapping with species lists or distributions improved the accuracy of local and global models, particularly for rare species. This procedure, which does not require model retraining, offers a practical means of enhancing model performance without additional computational overhead. In Table 1, we see that Bayesian remapping with species lists (which eliminates the possibility of predicting models not present in the test site) improves local/global models’ accuracy by 18.9% / 32.5% and 1% / 5.6% of Enonkishu and Kgalagadi, respectively. We see similar trends for remapping with species distributions and when using Bayesian remapping with a site-specific data source. In nearly all models, Bayesian remapping improved recall, indicating that performance on rare species can be improved. Since this remapping can be done without model retraining, it is a worthwhile procedure for practitioners to implement for improving trained model performance. However, this may not always be the right approach if data distribution shift (such as species range shift) is a possibility (see Section ??).

Effectiveness of Small Labeled Samples: Surpris-

Model	Training Data Source	Site knowledge (Bayes')		Accuracy		Avg. Recall	
		Spec. List	Spec. Freq	Enon.	Kgal.	Enon.	Kgal.
Shared Model	Local			0.353	0.494	0.181	0.099
			✓	0.334	0.615	0.187	0.111
		✓		0.542	0.504	0.198	0.123
	Global			0.177	0.459	0.076	0.115
			✓	0.270	0.650	0.124	0.113
		✓		0.502	0.550	0.153	0.265
Custom-trained Model	Local + Site			0.506	0.653	0.231	0.156
			✓	0.426	0.622	0.087	0.068
		✓		0.507	0.676	0.238	0.184
	Global + Site			0.434	0.567	0.212	0.147
			✓	0.423	0.647	0.088	0.093
		✓		0.474	0.620	0.269	0.292
	Site			0.759	0.879	0.288	0.262

Table 1. **Species identification performance on Kgalagadi (left) and Enonkishu (right) under different data sharing and capacity scenarios.** We define “Site” as data from the test site location; “Local” as the nearest site to your site of interest which for Kgalagdi is Karoo, and for Enonkishu is Serengeti; “Global” as all other available sites including the nearest site. We define “Spec. List” as access to the test site species list (without frequency) and “Spec. Freq.” as access to the relative frequency of each species in the data, both which require expert domain knowledge. We notice that as overall accuracy increases, the recall decreases. This is because as the model improves performance, it generally does better on common species. However, it decreases performance on rarer species, with many rare species having a 0% recall.

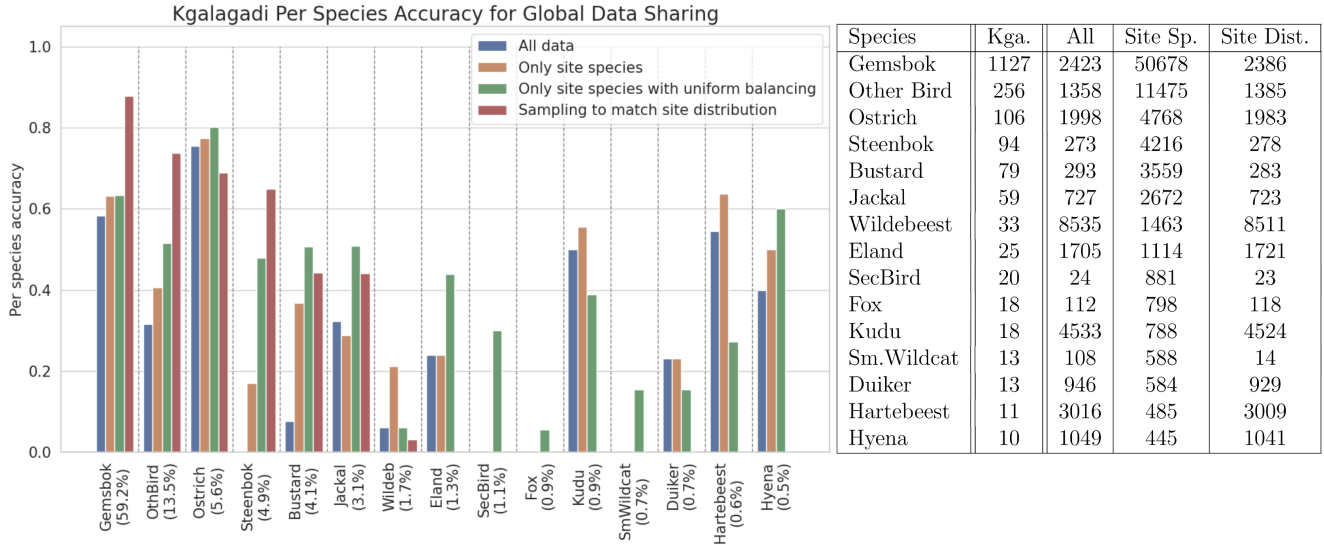


Figure 3. We compare sampling strategies with a fixed global data source and their effect on the downstream performance. The x-axis has the 15 common species in the training data, with relative percentages indicated. We find that using all data without any additional sampling procedure (indicated in blue) consistently performs poorly. Sampling to match the testing site distribution (indicated in red) often performs best for the more common species, but has a decreasing trend for rarer species, as the frequency of these examples decrease in the training set. Only using site species improves performance across classes, but to a lesser degree than uniform balancing and using only the site species (indicated in green), which improves performance significantly for rarer species. For example, uniform balancing is the only sampling procedure with any correct predictions for the Secretary Bird, Fox, and African Small Wildcat species but at a cost to performance on common species.

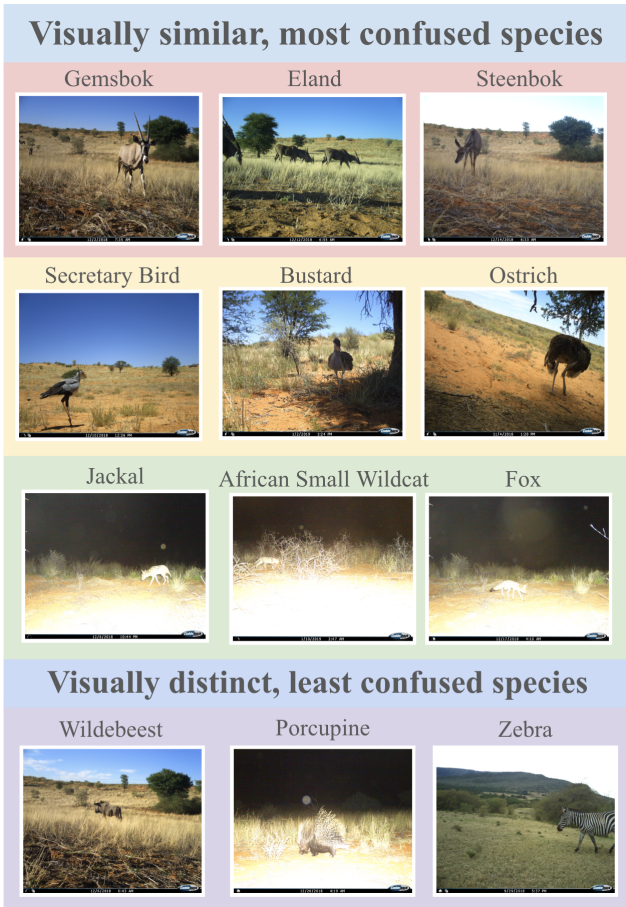


Figure 4. We display species that are visually similar and most confused for each other, indicating the harms in adding additional species/samples to a model. We also display visually distinct samples, which are rarely confused for other species and can be added to a training set without significantly decreasing performance.

ingly, our experiments demonstrate that a relatively small number of labeled site-specific samples can yield substantial improvements in model performance. For both Kgalagadi and Enonkishu, models trained exclusively on approximately ~ 1000 site-specific samples outperformed those trained on much larger global datasets ($\sim 100,000$ samples), emphasizing the importance of data quality over quantity. These results also show an increase in performance, invariant to dataset, size as training distribution of a shared model approaches that of the test distribution. Thus, when there is access to site data, it may not be necessary to expend resources to train on additional sources; in fact, adding other data can harm performance.

Sampling and Rarity: Our analysis reveals tradeoffs

between overall accuracy and average recall, particularly once one gets to custom models trained with rare species. Uniform sampling, as opposed to sampling according to test distribution, improved class performance for rare species (Appendix Table 2) but led to a slight drop in accuracy for common species. Uniform sampling with species list improved performance most on “median” species: not well-represented in test site but still above a prevalence threshold (usually > 50 images) within available shared data (Figure 3).

Visual similarity among species and background characteristics further influence model performance, particularly when training data includes species not seen at the test site. Consider the gemsbok antelope, which makes up over 59% of the Kgalagadi dataset. The gemsbok is confused for a steenbok 19% of the time when sampling to the relative species frequency or 37% of the time with uniform sampling (Figure 4). Likewise, jackals, African small wildcats, and foxes are often confused for one another due to similar background reflectance from nighttime flash. Once again, this highlights the importance of considering specific downstream use-case when selecting data and models.

G.1. Limitations of our case study

Our case study serves as an illustrative example of the intricate balance between data sharing and ML performance in ecological contexts. However, this is far from a comprehensive study:

Scope and Generalizability: Our study offers insights based on data from seven sites within a single continent. While informative, the findings may not be universally applicable to all ecological datasets or data-sharing scenarios. Notably, data from distant continents may exhibit even lower utility when applied to a specific site. Moreover, our focus on a single modality of ecological data limits the generalizability of our conclusions. Different sensor types, such as bioacoustic sensors or satellite imagery platforms, may introduce unique biases in species observation frequency.

Deployment Scenarios: Our case study primarily addresses a specific deployment scenario where users aim to label a season of collected data efficiently. While common, this scenario represents only one of several feasible deployment scenarios for ecologists. Variations in data splitting procedures, such as temporal or spatial splits for later seasons or new sites, could impact model performance differently. Furthermore, shifts in

data distribution over time or across sites could further affect model accuracy.

Image and Label Quality: We have not systematically explored the impact of image and label quality on model performance, presenting an additional dimension of complexity that warrants further investigation. While our study highlights overarching trends regarding category distribution per site and the value of data sharing, variations in data quality could influence model outcomes in practice.

H. Discussion and Recommendations

H.1. Reframing Discussions Around Data Value and Sharing in Ecology

The discourse on data sharing in ecology has predominantly centered on its direct ecological applications, overlooking its significance in building ML systems. Aligning data sharing priorities for different purposes or disentangling them could be instrumental. For instance, mechanisms that enable sharing species occurrence records rather than the images themselves for model training might offer a solution. However, careful consideration is necessary, as sharing rare species data to balance long-tailed distributions may introduce biases into species distribution models.

H.2. Need for Transparency Around Published Models

As ecological models become more accessible, transparency regarding the training data distribution becomes imperative. Datasheets and model cards serve as crucial mechanisms for increasing transparency, aiding organizations in assessing the suitability of models for their specific sites. This is particularly pertinent for ecological data with hierarchical structures, such as species types under a taxonomic tree.

H.3. Rare/Common Performance Tradeoffs Have Implications for Endangered Species Monitoring

Our study highlights tradeoffs between overall accuracy and average recall across species categories, influenced by factors like ML capacity, data sharing, and sampling decisions. These tradeoffs hold significant implications for endangered species monitoring, where maintaining accurate predictions for rare species is paramount. Balancing overall accuracy with the de-

tection of rare species necessitates careful consideration and resource allocation.

H.4. Incorporating Estimates of Distribution May Hinder the Detection of Species Range Shifts

While incorporating site-specific knowledge can enhance model performance, reliance on past estimates may hinder the detection of species range shifts due to climate change and land use changes. Relying solely on historical data may suppress our ability to detect species at range margins or during range shifts, posing challenges for ecological monitoring efforts.

H.5. Research directions to prioritize in ML

Data Attribution: Enhancing data sharing prioritization requires understanding the significance of individual data points. Model and data attribution methods offer insights into data valuation, yet they lack robustness under distribution shift and with long-tailed data.

Visual Similarity and Imbalanced Learning: ML models struggle with rare categories, especially in imbalanced datasets, exacerbating challenges with fine-grained categories. Developing methods to disambiguate fine-grained categories in long-tailed and shifting distributions is imperative to prevent systematic underprediction of rare species.

Data Augmentation via Generation: Generating synthetic data presents an alternative to balance long-tailed distributions and fill data gaps. While it has shown promise in improving rare class performance, generating diverse and high-fidelity data remains a challenge, particularly for fine-grained categories.

Federated Learning: Federated learning aims to train models across different systems without centralizing data. However, communication network speed and hardware variability pose challenges. Developing more flexible and lightweight methods is essential for resource-constrained ecological settings.

Human-AI Participation and Active Learning: Current models often require correction or verification before use in ecological applications. Research breakthroughs in enabling efficient participation between AI models and human experts through methods like selective prediction or active learning will facilitate broader impact, particularly in scenarios where specialized expertise is scarce.

I. Conclusions

Ecological applications inherently challenge the assumption of universal benefits from data sharing. Our case study illustrates the potential drawbacks, including unnecessary resource burdens and pressure to surrender data rights for minimal ML performance gains, particularly in traditionally undersampled areas with historic legacies of colonization. While data sharing can enhance ML model robustness and benefit the ecological community, it is essential to prioritize research enabling accurate data processing and informed decisions about data sharing.

J. Broader Impact Statement

Our work urges the ML community to adopt a nuanced understanding of data value and sharing in ecological contexts. By encouraging contextual considerations, we aim to enhance the utility of ML models for downstream users and initiate transparent discussions on data sharing prioritization. Our case study demonstrates the consequences of oversimplified approaches, emphasizing the importance of tailored solutions based on available resources and data. We acknowledge potential limitations outlined in Section G.1.

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K. Appendix

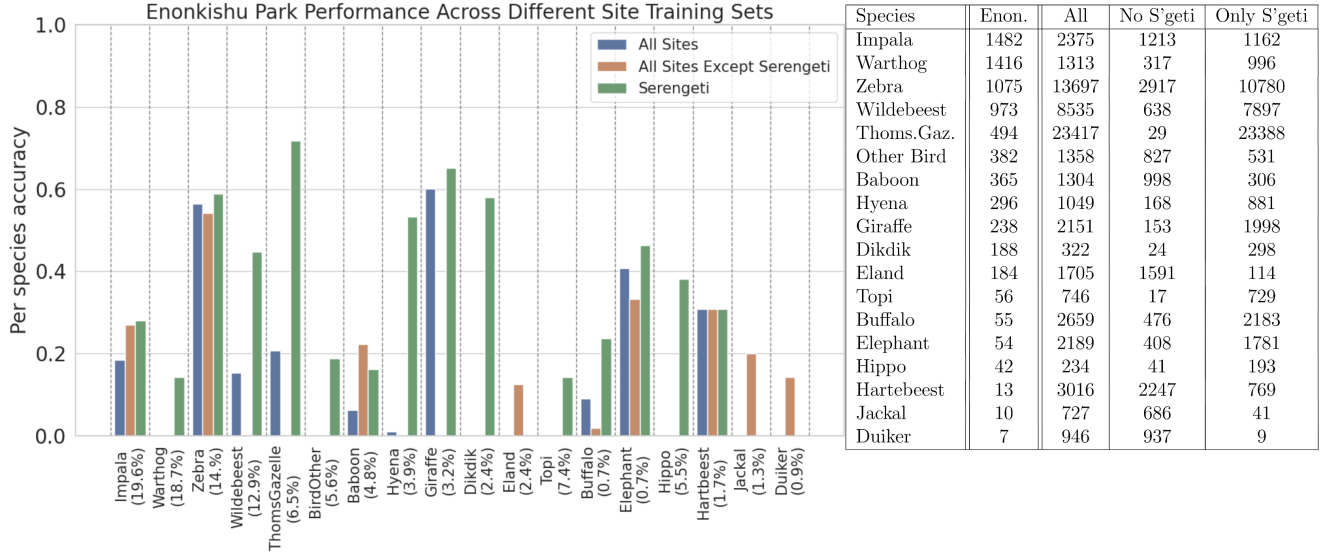


Figure 5. We compare different data sources and their effect on downstream species-level performance, comparing the inclusion of the nearest data-source Serengeti, which also makes up the majority of global data. (Left) We plot performance across the 15 most common species in the testing data, with relative percentages indicated. (Right) We indicate the image counts in the different variations of the data sources, sorted by test site frequency. We find that including different data sources changes performance depending on the species. Using just Serengeti performs best for the common species, where a local, nearby data source best matches test distribution and background image similarity, indicating that adding additional species/samples may confuse the model. Additionally, 9 out of 15 species are only predicted when the data source includes Serengeti (green and blue), indicating that these species are mostly present in the nearby data source. For example, the prevalence of Thompson’s Gazelle decreases from 23417 to 29 when excluding Serengeti. However, there are also several species that are only predicted when *excluding* Serengeti, including Eland, Jackal, and Duiker, which are suppressed with the addition of the large data source.

Training Data Source	Site Knowledge (Sampling)		Accuracy		Recall	
	Spec List	Spec Freq	Enon	Kgal	Enon	Kga
Local		✓	0.287	0.462	0.211	0.132
	✓		0.336	0.462	0.299	0.167
Global		✓	0.278	0.574	0.143	0.324
	✓		0.404	0.723	0.147	0.168
Local + Site		✓	0.503	0.719	0.211	0.131
	✓		0.475	0.669	0.369	0.408
Global + Site		✓	0.457	0.727	0.122	0.212
	✓		0.455	0.652	0.369	0.448
Site	✓	✓	0.759	0.879	0.288	0.262

Table 2. *Sampling ablations with knowledge*: We compare different sampling methods across data sources depending on access to site domain knowledge. As mentioned in Section 4, we under/oversample to the test distribution if we have access to species frequencies and uniformly sample if we have access to species lists. We find that adding these sampling setups consistently improves recall by eliminating species not of interest to our site. While results are mixed for which form of sampling is best, it is clear that either method improves performance for both testing sites by significant margins.